

Efficient Context-Aware Corpus-Grounded Query and Prompt Suggestion constrained by Creative Knowledge Graph

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Abstract

Creative design platforms support workflows such as template editing, social media creation, short-form video production, and AI-assisted asset generation. Users typically begin with short, underspecified queries and rely on the system to refine or expand their intent. Producing high-quality suggestions in this setting is challenging: outputs must preserve user intent, incorporate optional seasonal and regional context, and remain aligned with the underlying asset inventory. Traditional log-based approaches are not well suited for such structured, context-aware generation due to noisy, sparse, and unrepresentative query data.

We present a context-aware, corpus-grounded synthetic data framework for post-search query and generative prompt suggestion. Using 59M asset captions, we construct a two-stage offline pipeline that generates seed queries with semantic seasonality inference and expands them into related suggestions, narrow refinements, and generation-ready prompts. Generation is constrained via a Creative Knowledge Graph (CKG) and grounded using retrieval-augmented generation (RAG), increasing the proportion of high-relevance suggestions (score ≥ 5) by over 30%. The resulting dataset is used to fine-tune small language models through multi-task supervised fine-tuning with LoRA, enabling a unified model across all suggestion types.

Our architecture separates heavy offline synthesis from lightweight online inference. In production, a fine-tuned Gemma-3-4B model performs single-step generation without retrieval, reducing latency from seconds to 280ms using FP8 optimization. Evaluation combines LLM-as-a-judge scoring with search-execution grounding across relevance, diversity, seasonality, region alignment, and corpus consistency. These results demonstrate that synthetic supervision can effectively replace log-based training while delivering scalable, context-aware, and production-ready suggestion systems.

CCS Concepts

• **Information systems** → **Query suggestion; Retrieval models and ranking**; *Information retrieval*; • **Computing methodologies** → *Natural language generation; Machine learning*.

Keywords

Query suggestion, Generative prompt suggestion, Synthetic data generation, Context-aware search, Retrieval-augmented generation, Creative knowledge graph, Corpus-grounded generation, Parameter-efficient fine-tuning (PEFT), Small language models (SLMs), Multi-task learning

1 Introduction

Creative search systems increasingly support workflows such as asset discovery, template customization, and generative content

creation. Users typically begin with short, underspecified queries (e.g., "couple on beach") and expect the system to refine, expand, or transform that intent into actionable search queries or generation prompts. These suggestions play a critical role in the creative workflow: they help users discover complementary design possibilities that they may not have initially envisioned, explore alternative creative directions, and bridge the gap between vague intent and concrete, platform-actionable output. Designing a suggestion system that supports these diverse needs is a challenging problem.

Given a seed query and an asset type (e.g., image, template, video), a creative search system must provide multiple forms of assistance. Three types of suggestions are commonly required:

- **Related search suggestions**, which provide short alternative queries closely aligned with the original intent (e.g., "beach couple", "romantic beach scene").
- **Narrow refinements**, which add specificity while preserving the core meaning (e.g., "young couple walking on sunny beach", "romantic couple holding hands at sunset").
- **Generation-ready prompts**, which provide richer descriptive text suitable for direct use in generative models (e.g., "A romantic couple walking hand in hand along a serene beach at golden hour").

These suggestion types serve distinct user needs, namely rapid exploration, precise filtering, and content generation, and all must stem from the same underlying intent. A unified system capable of producing all three types from a single model offers significant architectural and operational advantages over maintaining separate pipelines for each.

Beyond intent preservation, real-world creative platforms introduce contextual constraints that further complicate suggestion generation. Suggestions may need to adapt to seasonal events (e.g., Valentine's Day or regional festivals) and geographic context, where cultural norms and visual motifs differ. For example, a query such as "couple on beach" during Valentine's Day may emphasize romantic lighting and heart-themed motifs, while the same query outside seasonal context should remain neutral. Similarly, culturally specific design elements may vary across regions even when the high-level theme is shared. Contextual adaptation must therefore be incorporated carefully without distorting the primary intent.

Traditional query suggestion systems rely heavily on historical user logs. While effective in many web search settings, direct reliance on user logs poses several limitations in creative platforms. User queries are often noisy, sparse, and ambiguous, especially for long-tail creative intents that lack sufficient interaction volume to produce reliable co-occurrence signals. Moreover, logs may contain sensitive behavioral signals and contextual information that cannot be directly reused for model training due to privacy and regulatory constraints. Finally, log data often underrepresents structured refinements and long-form generative prompts, which are increasingly important in modern creative workflows. These gaps make

it difficult to train models that can produce the full spectrum of suggestion types required by contemporary creative systems.

An additional requirement, often overlooked in existing work, is corpus alignment: suggestions must correspond to content that actually exists within the platform’s asset inventory. A suggestion that is semantically coherent but retrieves no relevant assets provides little practical value and may degrade user trust. Ensuring that generated suggestions remain grounded in the available corpus is therefore essential for production deployment.

These challenges collectively motivate the need for a unified, context-aware suggestion framework that supports multiple suggestion types while preserving intent, adapting to optional contextual signals, and remaining aligned with the available asset inventory. In this work, we address these requirements through a synthetic data generation pipeline constrained by a Creative Knowledge Graph and grounded via retrieval-augmented generation, coupled with a parameter-efficient fine-tuning strategy that enables efficient, low-latency inference in production. The following sections describe our approach and its empirical evaluation.

2 Contributions

The principal contributions of this work are as follows.

- **Seasonality-Aware Synthetic Supervision from Asset Captions.** We propose a two-stage synthetic data construction framework that replaces log-based supervision with corpus-derived signals obtained from a 59M-caption asset inventory. Seed queries and seasonality tags are inferred through LLM-based semantic reasoning, enabling structured supervision for related search suggestions, narrow refinements, and generation-ready prompts across diverse temporal and regional contexts.
- **Knowledge-Constrained, Retrieval-Grounded Data Generation.** We integrate Creative Knowledge Graph (CKG)-guided structured variation with retrieval-augmented grounding (RAG) during dataset construction. The CKG enforces asset-type-specific constraints spanning layout, motif, palette, composition, and tone, while RAG aligns generated outputs with the indexed corpus, improving grounding consistency of high-scoring suggestions by over 30% and mitigating semantic drift.
- **Unified Multi-Task Fine-Tuning with Efficient Deployment.** We jointly fine-tune instruction-tuned small language models via multi-task supervised fine-tuning (SFT) using a single LoRA adapter shared across all three suggestion types. By decoupling computationally intensive offline synthesis from lightweight online inference, the system supports single-step generation without retrieval at serving time.
- **Production-Scale Optimization and Evaluation Framework.** We optimize inference through LoRA weight merging and FP8 quantization, reducing end-to-end latency from seconds to 280 ms on commodity GPUs. We further introduce an automated evaluation pipeline that combines LLM-as-a-judge scoring with search-execution grounding to assess relevance, diversity, seasonality alignment, regional alignment, and corpus consistency at scale.

While the individual techniques used in our system are not novel in isolation, our contribution lies in their integration into a unified, seasonality-aware synthetic data pipeline tailored for creative search and generation tasks at scale. To the best of our knowledge, prior work has not combined seasonality-aware query synthesis, asset-type-specific knowledge constraints, and corpus-grounded generation within a single production-oriented pipeline. This integration enables structured, context-aware supervision that supports multiple suggestion types while maintaining corpus alignment and low-latency deployment.

3 Related Work

Our work draws upon and extends research across several interconnected areas: query suggestion, generative query reformulation, synthetic supervision, retrieval-augmented generation, and parameter-efficient adaptation of large language models.

Query Suggestion and Reformulation. Traditional query suggestion methods rely on query logs, click-through data, and session-based modeling to generate refinements and completions. Earlier approaches employed probabilistic language models, graph-based methods, and sequence-to-sequence architectures to predict likely successor queries from historical interaction logs [6]. While effective at scale, these systems depend on large volumes of behavioral data and do not explicitly model contextual signals such as seasonality or cultural variation, limiting their applicability in creative search settings where such signals are essential. Recent research has explored neural and generative approaches to query recommendation. Min et al. [10] propose a generative framework that aligns LLM-based query recommendation with user preference signals. Baek et al. [5] introduce knowledge-augmented LLMs for personalized contextual query suggestion, demonstrating the value of external knowledge integration. Chen et al. [7] investigate multi-stage LLM-based query expansion strategies for retrieval tasks, highlighting the potential of iterative refinement. Multimodal query suggestion has been explored in image retrieval settings [13, 14], extending the paradigm beyond text-only interactions. In contrast to log-dependent or personalization-heavy approaches, our work focuses on seasonality-aware, corpus-grounded synthetic supervision that eliminates the need for access to sensitive user interaction logs while explicitly modeling temporal and regional context.

Synthetic Data for Retrieval and Generation. Synthetic data generation has gained considerable attention as a means of addressing data sparsity, domain specificity, and privacy constraints. Comprehensive surveys highlight the rapidly expanding use of LLMs for synthetic supervision across diverse domains [2]. Recent work demonstrates that synthetic query generation can effectively enhance retrieval model performance in low-resource and domain-specific settings [4], suggesting that carefully constructed synthetic signals can serve as viable alternatives to human-generated or log-derived training data. Our approach extends this paradigm by incorporating structured seasonality inference, Creative Knowledge Graph constraints, and retrieval-augmented grounding to ensure that generated outputs remain aligned with the platform’s asset corpus—addressing a gap in prior synthetic data pipelines that typically lack such domain-specific structural and inventory constraints.

Retrieval-Augmented Generation (RAG). Retrieval-augmented generation combines parametric language models with non-parametric retrieval to improve factual grounding and mitigate hallucination [9]. While RAG is conventionally deployed at inference time to provide models with dynamic external context, we apply retrieval during offline dataset construction rather than at serving time. This design choice ensures that synthetic suggestions are empirically supported by real asset captions in the indexed corpus, thereby improving grounding consistency while preserving the lightweight, retrieval-free inference path required for production latency constraints.

Parameter-Efficient Fine-Tuning. Low-Rank Adaptation (LoRA) [8] enables efficient fine-tuning of large language models by injecting trainable low-rank decomposition matrices into transformer layers, substantially reducing the number of trainable parameters while preserving model expressiveness. LoRA has become a widely adopted technique for adapting large pretrained models under limited compute budgets. We leverage LoRA to fine-tune Gemma-family small language models in a multi-task configuration, sharing a single adapter across all three suggestion types to enable scalable training and efficient deployment without maintaining separate model instances.

LLM-Based Evaluation. Evaluating open-ended generation tasks such as query suggestion presents inherent challenges, as traditional metrics (e.g., BLEU, ROUGE) often fail to capture semantic adequacy and contextual appropriateness. Recent work has investigated the use of LLMs as automated evaluators—commonly referred to as the “LLM-as-a-judge” paradigm—for scoring generation quality across multiple dimensions [3, 15]. We adopt this approach and extend it with search-execution grounding, wherein generated queries are executed against the production search index and evaluated for retrieval relevance. This dual evaluation strategy provides complementary signals: LLM-based scoring captures semantic quality, while execution-based grounding measures practical utility within the deployed system.

Positioning. Unlike prior log-based or personalization-driven approaches, our system uniquely integrates seasonality-aware synthetic supervision, Creative Knowledge Graph-guided structured variation, and retrieval-grounded dataset construction within a unified pipeline, followed by parameter-efficient SLM fine-tuning optimized for production deployment. This combination enables privacy-preserving, context-aware query and prompt suggestion with strong corpus alignment and low-latency serving—capabilities that, to the best of our knowledge, have not been jointly addressed in prior work.

4 Dataset Construction

This section describes the end-to-end construction of our seasonality-aware synthetic training dataset. The pipeline is illustrated in Figure 1. Our objective is to construct high-quality, context-aware supervision without relying on user search logs, while ensuring corpus grounding and production relevance.

4.1 Overview of the Pipeline

Dataset construction follows a two-stage generation process performed entirely offline:

- (1) Caption $\rightarrow$ Seed Query + Seasonality Inference
- (2) Seed Query $\rightarrow$ Context-Aware Suggestions and Generative Prompts

The full caption corpus is used for retrieval grounding, while a sampled subset is used to generate structured training examples. The resulting dataset is later used to fine-tune a small language model for production deployment.

4.2 Caption Corpus Preparation

We begin with approximately 59 million publicly available asset captions spanning images, videos, templates, backgrounds, icons, and other creative assets. These captions were generated using a vision-language model and describe assets that are already available in the platform.

Before indexing, captions are preprocessed by:

- Removing empty or malformed entries,
- Cleaning HTML artifacts and non-textual noise,
- Filtering profanity,
- Retaining captions shorter than 30 words.

No fuzzy deduplication is applied. The cleaned 59M captions are embedded using nomic embeddings [12] and indexed in a vector store for retrieval-augmented generation.

From this corpus, we sample 500,000 captions to generate synthetic training examples. Sampling ensures coverage across asset types, and creative domains while keeping generation computationally feasible.

4.3 Stage 1: Seed Query and Seasonality Inference

In the first stage, each sampled caption is converted into a minimal, intent-faithful seed query using an instruction-tuned large language model. The goal is to simulate realistic user search behavior while preserving alignment with the underlying asset.

During this step, the model also performs semantic seasonality inference. Instead of keyword matching or metadata lookup, seasonality is assigned only when strongly implied by caption semantics. For example, pumpkins with dark styling may trigger a Halloween tag, while festive decorative lights may trigger a festival tag. If no clear seasonal signal exists, seasonality remains unset.

The system supports approximately 170 distinct seasonality tags covering global festivals, cultural events, and recurring occasions.

4.4 Region Derivation

Region is not directly inferred from captions. Instead, region is derived from a predefined season-to-region compatibility mapping. When seasonality is present, region is inferred accordingly.

4.5 Stage 2: Context-Aware Suggestion and Prompt Generation

In the second stage, each seed query is expanded into structured outputs using large language models. Generation is conditioned on the seed query, asset type, inferred seasonality, and derived region (when applicable). This conditioning enables context-aware refinement while preserving the core search intent.

For each seed query, we generate three categories of outputs:

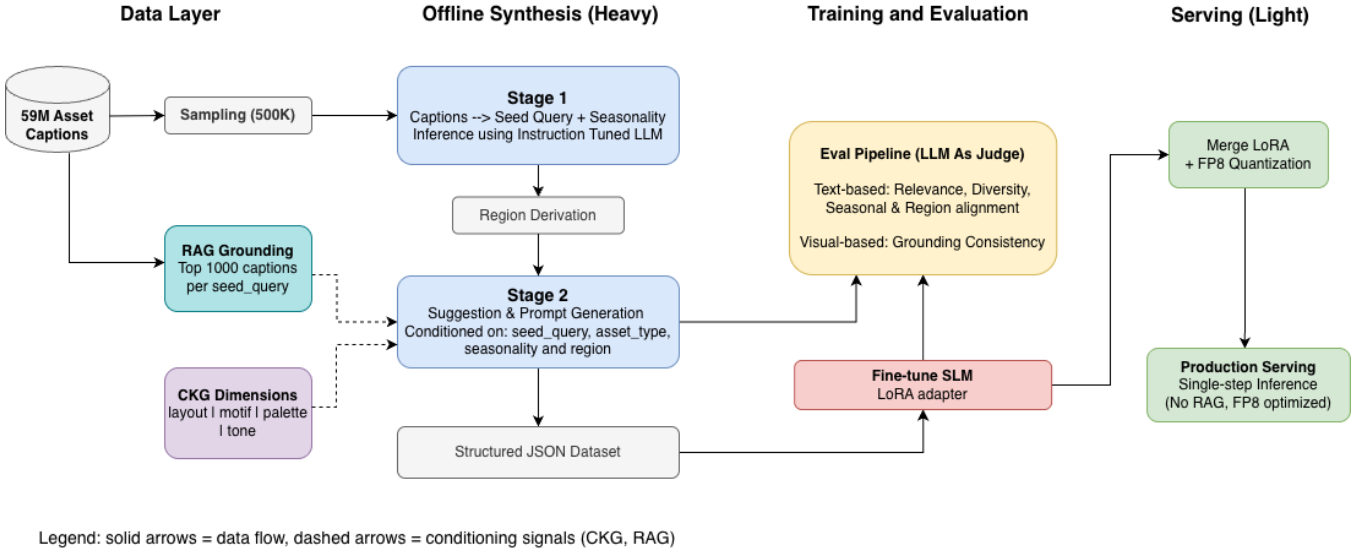


Figure 1: End-to-end seasonality-aware corpus-grounded synthetic data and serving pipeline. The Data Layer (left) consists of 59M asset captions used to generate seed queries via sampling. The Offline Synthesis stage (center) performs seed query and seasonality inference (Stage 1), region derivation, and context-aware suggestion and prompt generation (Stage 2) conditioned on seed query, asset type, seasonality, and region. Creative Knowledge Graph (CKG) dimensions and retrieval-augmented grounding (RAG) provide structured conditioning signals during generation. The resulting structured JSON dataset is used to fine-tune a small language model. The Training & Serving stage (right) includes LoRA-based fine-tuning, LLM-based evaluation, LoRA merging with FP8 optimization, and lightweight single-step production inference without RAG. Solid arrows denote data flow, while dashed arrows denote conditioning signals.

- 10–15 related search suggestions,
- 5–10 narrow refinements,
- 3–5 generation-ready prompts suitable for asset creation.

Importantly, during dataset construction we allow variability in the number of suggestions and their length. This design enables the downstream small language model to learn flexible generation behavior, allowing it to produce any number of suggestions with varying word lengths at runtime. This flexibility is critical because the system serves multiple product surfaces (e.g., search, template editors, and generative workflows), each with different constraints.

Seasonality conditioning is applied only when strongly supported by caption semantics inferred during Stage 1. Negative seasonality conditioning is excluded during data generation, as early experiments showed that explicit negative signals increased hallucination and intent drift. Instead, negative seasonality robustness is introduced later during model training.

Region names are never explicitly included in generated outputs; instead, regional context is reflected implicitly through culturally appropriate motifs, styling, and visual elements. In our system, region corresponds to the user’s geographic location (country). When a query is issued, suggestions are adapted to reflect cultural norms and design conventions commonly associated with the region.

Seasonality and region influence suggestions in complementary ways. Seasonality modifies the temporal theme of a suggestion (e.g., a “cake poster” during Valentine’s Day may incorporate heart motifs and soft pink palettes). Region, by contrast, influences how that theme is expressed culturally. For example, a festival-themed poster

generated for users in India may emphasize traditional decorative elements, specific color harmonies, or culturally familiar symbols, whereas the same seasonal theme for users in another country may use different stylistic conventions.

Region is inferred indirectly via a season-to-region compatibility mapping or user context, but region names themselves are never surfaced in the suggestion text. The challenge is to adapt suggestions to the user’s cultural environment without explicitly referencing geographic labels, while preserving the primary intent of the seed query.

To control variation and prevent semantic drift, we incorporate Creative Knowledge Graph (CKG) constraints during generation. Without structured constraints, early experiments showed that large language models tend to produce near-synonymous suggestions, overly generic refinements, or suggestions that subtly shift the original intent (e.g., drifting from a design task to unrelated object categories). In some cases, the model introduced unsupported stylistic attributes that were not aligned with the asset inventory, increasing hallucination and reducing grounding consistency.

The CKG provides asset-type-specific creative dimensions that explicitly define how suggestions may vary. These dimensions differ across asset types. For example, for *template* assets, relevant CKG nodes include layout (e.g., centered, grid), color palette (e.g., pastel, monochrome), visual motifs (e.g., floral, geometric), orientation (portrait, landscape), and tone (formal, festive). For *image* assets, nodes include subject attributes, scene context (indoor, outdoor), camera composition (close-up, wide shot), lighting, and mood.

During generation, variations are explicitly guided across these nodes rather than allowing unconstrained free-form expansion. Short related suggestions typically vary one to two dimensions (e.g., motif + palette), while longer refinements and generative prompts vary two to four dimensions (e.g., layout + motif + palette + tone). This structured variation improves diversity while preserving the core intent of the seed query.

Each generated output includes the varied CKG nodes and a short reasoning explanation. This explicit annotation enables traceability, supports automated evaluation, and allows systematic control over diversity across creative dimensions.

Model Exploration for Dataset Generation. We explored multiple large language models for Stage 2 generation, including Llama-4-Scout [1] and Llama-3.1-Nemotron-Ultra-253B [11]. For each model, we generated synthetic samples under identical prompting and conditioning settings. The resulting datasets were evaluated using our automated LLM-based evaluation pipeline, which measures relevance, diversity, seasonality alignment, region alignment, and grounding consistency. Based on evaluation results across all major metrics, the dataset generated using the Nemotron Ultra model demonstrated stronger context alignment, better seasonality consistency, and higher grounding validity compared to the Llama-4-Scout outputs. Therefore, the Nemotron Ultra-generated dataset was selected for downstream small model training.

4.6 Retrieval-Augmented Grounding

To ensure suggestions remain aligned with the available asset inventory, we apply retrieval-augmented generation (RAG) during dataset construction. All 59M captions are embedded using a Nomic embedding model and indexed in a vector store. For each seed query, the top semantically similar captions are retrieved and appended as grounding context during generation. This grounding mechanism improves the proportion of high-scoring suggestions by over 30% compared to non-grounded generation and substantially reduces unsupported outputs. Retrieval is used only during offline dataset construction and is not required during production inference.

4.7 Structured Dataset and Offline-Online Separation

The final dataset consists of structured JSON entries containing:

- Seed query,
- Asset type,
- Seasonality (optional),
- Region (optional),
- Related suggestions,
- Narrow refinements,
- Generation-ready prompts,
- Varied CKG nodes and reasoning annotations.

The multi-step LLM pipeline—seed generation, seasonality inference, region derivation, retrieval grounding, and knowledge-constrained expansion—is used exclusively offline to construct the synthetic dataset.

Using this dataset, we fine-tune lightweight small language models for deployment. In production, only the trained small language model is served, enabling efficient, single-step, seasonality-aware

query and prompt generation without external retrieval or large-model dependencies.

5 Training Details

We fine-tune instruction-tuned chat models using supervised fine-tuning (SFT) under a chat-completion objective. Our primary production model is google/gemma-3-4b-it. All tasks are trained using a single shared LoRA adapter.

5.1 Multi-Task Chat-Based Supervised Fine-Tuning

The synthetic dataset consists of three task types, with example instruction–input–response samples provided in Table 4.

- **Related Search Suggestions**
- **Narrow Search Refinements**
- **Generative Result Prompts**

All three tasks are trained jointly using a single LoRA adapter to encourage shared contextual reasoning across suggestion types. Each training example is formatted as a chat-style instruction–response pair consisting of:

- A system instruction describing the generation behavior,
- A structured user input containing seed_query, asset_type, seasonality, and region,
- A target assistant response in structured JSON format.

We adopt a completion-only training strategy where the loss is computed only over assistant tokens. This enables the model to learn structured JSON outputs while preserving instruction-following behavior.

5.2 Instruction Diversity for Generalization

To improve robustness and avoid overfitting to a single instruction template, we generate multiple variations of prompts for each task type. Specifically, we use approximately 20 distinct instruction formulations per task.

This design encourages the model to generalize beyond fixed phrasing and leverage the pretrained model’s open-domain knowledge. Instruction diversity improves stability across surfaces and reduces prompt sensitivity during inference.

5.3 LoRA Configuration

We apply Low-Rank Adaptation (LoRA) to adapt the base model efficiently:

- Rank (r): 16
- LoRA alpha: 32
- LoRA dropout: 0.2
- Target modules: q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj

All three tasks are trained under a unified LoRA configuration, allowing shared semantic reasoning across suggestion generation and generative prompting.

5.4 Optimization Setup

Key training hyperparameters for the 4B model are:

- Epochs: 15

- Learning rate: 4×10^{-5}
- Weight decay: 0.05
- Scheduler: Linear with 10% warmup
- Per-device batch size: 4
- Gradient accumulation steps: 4
- Max sequence length: 1536
- Precision: bfloat16 (bf16)

The best checkpoint is selected based on validation loss and downstream LLM-based evaluation metrics.

6 Evaluation

Evaluating post-search query suggestions and generative prompts is challenging due to the absence of explicit ground-truth labels. We therefore design an automated LLM-based evaluation framework that measures quality across multiple complementary dimensions.

6.1 Evaluation Setup

We evaluate on a held-out validation set of approximately 30K user queries sampled to reflect real-world query distributions across head, torso, and tail segments. The dataset is further stratified across different months and regions to ensure coverage of seasonal and regional variations.

We compare fine-tuned Gemma-3-1B and Gemma-3-4B models under identical decoding settings. Evaluation uses a larger LLM (Qwen/Qwen2.5-32B) as an automated evaluator (“LLM-as-a-judge”) to score each generated suggestion or prompt across four metrics: relevance, diversity, seasonality alignment, and region alignment. Grounding consistency is assessed separately via a search-execution check (Section 8.1), in which generated queries are executed against the production index and the retrieved assets are judged for relevance.

6.2 Evaluation Metrics

We evaluate the following metrics and score them from 0 to 5 (worst to best).

Relevance. Measures how well the generated suggestion preserves the core intent of the seed query along with the specified asset type and task type. The evaluator judges whether the suggestion is a usable, intent-aligned refinement for asset search.

Diversity. Measures internal variety among the set of suggestions for a given seed (e.g., semantic spread, differing creative dimensions). High diversity indicates the model proposes multiple distinct, useful directions rather than near-synonyms.

Seasonality Alignment. Measures whether outputs correctly reflect seasonal context when seasonality is applicable. If seasonality is absent, unnecessary seasonal content is penalized.

Region Alignment. Measures cultural appropriateness and contextual consistency when region is inferred. The evaluator checks for implicitly appropriate motifs, styling, or objects that reflect regionally compatible cues without explicit region names.

Grounding Consistency. Grounding consistency measures whether a generated suggestion is supported by the actual asset inventory.

For each generated query, we execute the query against the production search index and retrieve the top results. A larger LLM evaluates whether the retrieved assets are relevant and semantically aligned with the suggested query. If the retrieved results are relevant, the suggestion is considered grounded. This metric directly measures practical search alignment rather than relying solely on embedding similarity.

6.3 Results: Gemma-3-1B vs Gemma-3-4B

Table 1 compares Gemma-3-1B and Gemma-3-4B across the four judge-scored evaluation metrics.

Table 1: Comparison of Gemma-3-1B and Gemma-3-4B on validation set (higher is better).

Model	Relevance	Diversity	Seasonality	Region
Gemma-3-1B	3.50	3.20	2.23	3.79
Gemma-3-4B	3.69	4.02	4.56	4.41

Gemma-3-4B outperforms Gemma-3-1B across all four metrics. The most pronounced gains are observed in seasonality alignment, indicating that the larger model better captures contextual seasonal signals and produces outputs more consistent with the corpus. Diversity also improves with model scale, showing that Gemma-3-4B proposes a wider range of distinct suggestion directions rather than near-synonyms. Gemma-3-1B remains useful for low-latency or resource-constrained deployments but exhibits weaker seasonality adaptation and lower suggestion diversity.

6.4 Human Evaluation

To complement automated evaluation, we conduct human studies to assess the quality of generated suggestions.

Comparison with Log-Based System. We first compare our model against a traditional log-based query suggestion system. The evaluation is conducted on 400 queries sampled across asset types and query distributions. Approximately 20 human annotators compare suggestions from both systems and select the preferred output based on relevance, usefulness, and contextual alignment. Annotators are also allowed to indicate ties or poor-quality outputs.

The results are as follows:

- **72%:** our model is preferred,
- **11%:** log-based system is preferred,
- **16%:** both are equally preferred,
- **1%:** both are considered poor.

These results demonstrate a clear preference for the proposed approach over traditional log-based suggestions.

Absolute Quality Assessment. We further evaluate the model at scale using 10,000 queries, where human annotators rate generated suggestions on a 1–5 scale based on overall quality, relevance, and usefulness.

The model achieves an average rating of **4.18**, indicating consistently high-quality suggestions across a diverse set of queries.

Discussion. Together, these evaluations show that the proposed synthetic data-driven approach not only outperforms traditional systems in pairwise comparisons but also achieves high absolute quality in human judgment, validating its effectiveness in real-world settings.

7 Inference Optimization and Benchmarking

This section describes inference optimization strategies and benchmarking results for the production model. While dataset construction involves multi-step LLM pipelines and retrieval, production inference uses a single fine-tuned small language model (Gemma-3-4B with LoRA merged), enabling efficient one-step generation.

7.1 Serving Setup

All benchmarks were conducted using google/gemma-3-4b-it fine-tuned with LoRA. For production deployment, LoRA adapters were merged into the base model to reduce inference overhead.

Inference was performed using vLLM on commodity GPUs. We evaluated both BF16 and FP8 configurations under identical decoding parameters.

7.2 Latency and Throughput

We measure average latency, P95 latency, and requests per second (req/s) over 50 sequential requests for related search suggestion workloads.

Table 2: Inference performance comparison (Gemma-3-4B).

Configuration	Avg (ms)	P95 (ms)	Req/s
BF16 + LoRA (vLLM 0.10.0)	428.1	490.0	2.38
BF16 + LoRA (vLLM 0.13.0)	365.6	365.6	2.74
BF16 + LoRA (vLLM 0.14.1)	347.7	414.7	2.88
FP8 + LoRA (vLLM 0.14.1)	289.3	343.7	3.46
FP8 + LoRA Merged (vLLM 0.14.1)	238.1	269.7	4.20

Upgrading vLLM versions significantly improved performance. Moving from vLLM 0.10.0 to 0.14.1 reduced average latency by about 20% under BF16. FP8 quantization further reduced latency by approximately 17% compared to BF16 under the same version. Additionally, merging LoRA adapters into the base model provided additional latency reduction of about 17% by eliminating adapter overhead.

8 Ablation Studies

To understand the contribution of each design choice, we perform controlled ablation studies on dataset construction, model scale, parameter-efficient tuning, and decoding configuration.

8.1 Effect of Retrieval-Augmented Grounding

We compare evaluation metrics on synthetic data generated with and without retrieval-augmented grounding (RAG). In the non-RAG setup, seed queries are expanded without incorporating corpus evidence, whereas in the RAG setup, top-K relevant captions from the 59M asset corpus are retrieved and used as grounding context during generation.

Across the LLM-based evaluation metrics in Table 3, we observe moderate improvements in relevance, diversity, and seasonality alignment. In addition to these metrics, we evaluate grounding consistency using a search-execution pipeline, where generated suggestions are executed against the production search index and the retrieved assets are assessed for relevance.

Under this evaluation, RAG improves grounding consistency significantly, increasing the proportion of suggestions that retrieve relevant assets (score ≥ 5) by approximately 30%. This indicates that incorporating corpus evidence during data generation reduces hallucination and produces suggestions that are more reliably aligned with the underlying asset inventory.

Table 3: Impact of RAG on validation metrics.

Setting	Relevance	Diversity	Seasonality	Region
No RAG	3.55	3.07	4.46	4.36
With RAG	3.65	3.38	4.80	4.39

8.2 Model Scale: Gemma-1B vs Gemma-4B

We compare Gemma-3-1B and Gemma-3-4B under identical training and decoding settings, as shown in Table 1. The 4B model consistently outperforms the 1B model across all metrics, with the largest gains observed in seasonality and region alignment. The increased model capacity improves contextual reasoning and reduces semantic drift in generated suggestions.

Additionally, the 4B model demonstrates stronger generalization to unseen or long-tail queries by leveraging its broader pretrained knowledge. This is particularly important in creative search, where user queries are often diverse and not well represented in training data. Based on these improvements, we select Gemma-3-4B as the production model.

8.3 LoRA Rank Sensitivity

We evaluated LoRA ranks of 8, 16, and 32 on the 4B model while keeping all other hyperparameters fixed. Lower ranks ($r=8$) resulted in slightly reduced diversity and seasonality alignment. Higher ranks ($r=32$) showed marginal gains in relevance but increased overfitting and training instability. Rank 16 provided the best balance between generalization and performance, and was selected for final training.

8.4 Decoding Hyperparameters

We evaluated combinations of temperature $\{0.5, 0.7, 0.9, 1.1, 1.2\}$ and repetition penalty $\{1.0, 1.1, 1.2\}$ for both related and narrow suggestion tasks as shown in figure 2. Lower temperatures (0.5–0.7) produced more relevant but less diverse suggestions. Higher repetition penalties increased diversity among suggestions but slightly reduced relevance. Across both related and narrow suggestion tasks, the configuration:

Temperature = 0.5, Repetition Penalty = 1.1

provided the best trade-off between relevance and diversity, while maintaining strong seasonality and region alignment. This configuration was selected for production deployment.

8.5 Effect of CKG-Constrained Generation

To analyze the role of Creative Knowledge Graph (CKG) constraints, we compare synthetic data generated with and without CKG-guided variation during dataset construction.

Without CKG constraints, the generation process tends to produce near-synonymous suggestions with limited variation across creative attributes. In many cases, suggestions drift toward generic reformulations of the seed query, failing to capture asset-specific properties such as layout, composition, or stylistic intent. For example, for a query such as “pumpkin poster”, unconstrained generation often produces variations like “pumpkin flyer”, “pumpkin design”, or “autumn poster”, which exhibit lexical diversity but lack meaningful variation across design dimensions.

In contrast, CKG-guided generation explicitly enforces variation across structured creative dimensions such as layout, motif, palette, composition, and tone. This results in suggestions that reflect distinct design interpretations (e.g., “rustic pumpkin harvest poster”, “minimalist autumn flyer with warm palette”, “grid-based festival poster with pumpkin motifs”), thereby increasing diversity in a controlled manner.

We also observe that without CKG constraints, the model is more prone to semantic drift, occasionally introducing attributes that are not aligned with the intended asset type or design context. CKG conditioning mitigates this issue by restricting generation to valid combinations of asset-type-specific attributes.

Overall, these observations indicate that CKG constraints play a critical role in enabling structured diversity, reducing semantic drift, and ensuring that generated suggestions remain consistent with the intended creative dimensions.

9 Conclusion

We presented a context-aware, corpus-grounded framework for post-search query and generative prompt suggestion that replaces noisy user logs with structured synthetic supervision. By leveraging a large corpus of 59M asset captions, we construct a two-stage offline data generation pipeline that infers seed queries, seasonality, and region, and expands them into related suggestions, narrow refinements, and generation-ready prompts under Creative Knowledge Graph (CKG) constraints.

Retrieval-augmented grounding ensures alignment with the underlying asset inventory, improving grounding consistency of high-scoring suggestions by over 30%. Structured CKG-guided variation enables controlled diversity while preserving core intent. The resulting synthetic dataset supports effective fine-tuning of lightweight small language models.

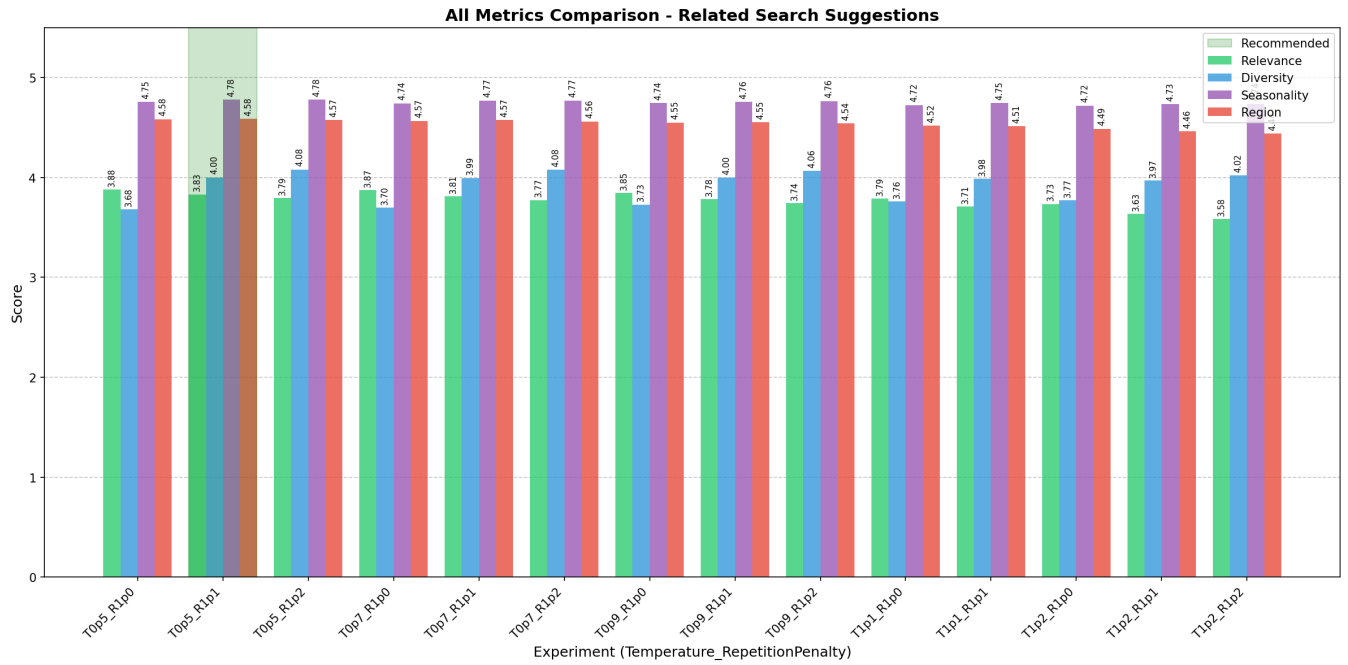
A key architectural contribution is the separation of heavy offline synthesis from lightweight online inference. While dataset construction involves large models, retrieval pipelines, and structured reasoning, production deployment uses a single fine-tuned small language model operating in one step. With LoRA adaptation and FP8 inference optimization, we reduce latency from seconds to approximately 280ms while maintaining strong seasonality alignment and contextual quality.

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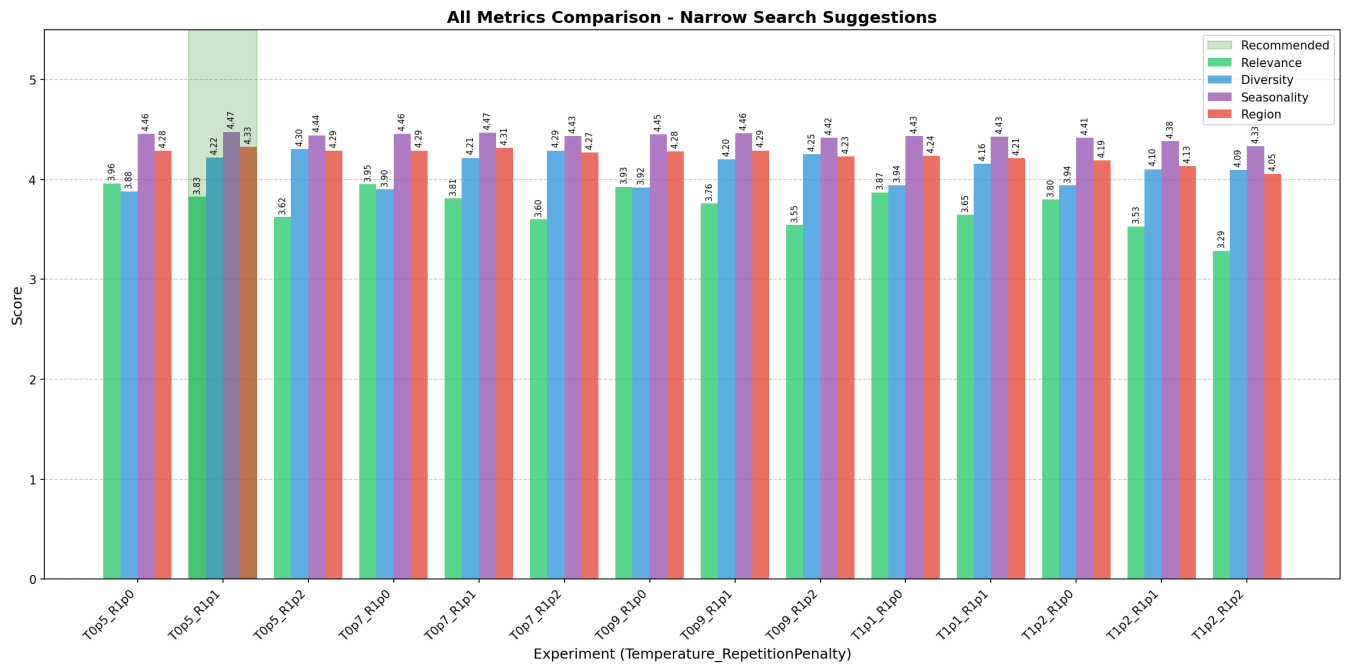
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Table 4: Example supervised fine-tuning samples (Task → Instruction | Input | Response).

Task	Instruction	Input	Response
Related Search	Instruction: You are trained to generate helpful creative search suggestions across global user bases. Given a user query, asset category, and optional context (season/region), return 10 possible search phrases that are short (4–5 words) and aligned with intent. Output JSON: {"related_search_suggestions": [...]}	Input: {seed_query: "summer sunset hiking pacific ocean landscape", asset_type: "image", seasonality: "breast_cancer_awareness_month", region: "United States"}	Response: {"related_search_suggestions": ["hiking trail pacific coast sunset", "summer mountain hike ocean view", "pacific ocean sunset adventure", "coastal hike sunset panorama", "summer wilderness sunset exploration", "pacific trail sunset hike", "ocean view mountain hike sunset", "summer coastal path sunset", "hiker silhouette pacific sunset", "wilderness adventure sunset landscape"]}
Narrow Search	Instruction: Act as a creative query expert. Analyze the user’s seed query and asset type (plus optional seasonality/region). Generate 5 narrow, specific search queries (11–15 words). Use seasonality/region only if implied. Output JSON: {"narrow_search_suggestions": [...]}	Input: {seed_query: "school supply drive poster", asset_type: "template", seasonality: "raksha_bandhan", region: "India"}	Response: {"narrow_search_suggestions": ["colorful school supply drive poster template with donation details and educational icons", "back to school supply drive poster with bold fonts and vibrant background for charity event", "minimalist school supply drive template with space for text and school-themed illustrations", "school supply donation poster with green and blue color scheme and checklist layout", "customizable school supply drive template with photo placeholders and bold headlines"]}
Generative Prompt	Instruction: You are a creative prompting system for generative AI-based asset generation. Given the user’s query and context (asset_type, seasonality, region), craft 3 descriptive prompts (32–44 words). Output JSON: {"generate_result_prompt": [...]}	Input: {seed_query: "relaxing cafe video, woman reading", asset_type: "video", seasonality: "passover", region: "United Kingdom"}	Response: {"generate_result_prompt": ["A serene video of a woman in her late 20s sitting in a cozy cafe with large windows, softly lit by natural light, reading a novel with a cup of steaming coffee beside her, captured in slow motion with a shallow depth of field.", "Close-up shot of a woman’s hands turning the pages of a book, with a blurred background of a bustling cafe, warm lighting, and a subtle smile on her face as she reads.", "Aerial view of a woman sitting alone at a small round table in a quaint outdoor cafe, surrounded by greenery, reading a book with a cup of coffee, filmed in 4K with a drone for a unique perspective."]}



(a) Related search suggestions



(b) Narrow search suggestions

Figure 2: Extended hyperparameter tuning results for decoding configurations across temperature and repetition penalty. These plots provide a detailed comparison across relevance, diversity, seasonality alignment, and region alignment metrics.